LEVEL 6 Topic 7

LIGHTNING PROTECTION.

Defination of terms.

1. **Lightening / Lightening flash or discharge**
- The discharge of electrical Currents in the air / atmosphere Caused by two Charged Clouds.

2. **Lightening density.**
- The frequency by which a Specific type of Lightening flashes / discharges in an area over a given period of time.

3. **Lightening Strike.**
- Action or effect Caused by a Lightening discharge.

4. **Lightening Strike attachment Point.**
- The Contact Point where the Lightening Strike touches the ground or a Structure.

5. **Lightening Protection System.**
- A Combination of Components designed from Conductors that are installed to reduce the effect of Lightening.

6. **Zone of Protection**
- An area protected from Lightning by the Protection System.

Types of Lightning Strikes.

- **Direct Strike**
- Occurs when Lightning Strikes an individual o Organism.
- **Side flash**
- Occurs when Lightning Strikes a Structure and then the Strike extends to individual Organism.

- **Ground Current**

  - A form of dissipated Strike that results from objects discharge being directed to the ground. The dissipated energy can possibly electrocute individuals/organisms nearby.

- **Conduction**

  - Occurs in case an individual/organism comes into contact with metal material that has been struck by lightning.

Components of Lightning Protection System.

The basic components of a lightning protection system include;-

(i) **Air terminals**

- These are metal rods that serve as the first exposure point for lightning strikes. They are set up on the roof the structure.

(ii) **Main Conductor Cables.**

- Aluminium Copper Cable which is connected to the air terminal and runs/passes on roof edges of adjacent buildings high points and is routed to the ground.

(iii) **Ground rods.**

- Composed of a corrosion resistant alloy of copper and steel and are connected to the main conductor cables. Ground rods are part of the earthing system.

(iv) **Bonding Conductor**

- An equal potential conductor which connects the main conductor/grounding conductor and the earthing rods/system.

(i) Lightning arresters | Surge protection

- Electrical devices installed in or on a building's electrical components and designed to protect the equipment from the electrical surge that occurs when lightning strikes a nearby power line.

Types of Lightning Protection System.

1. Franklin Rods.

- ~~These are terminals of~~ These are metal rods installed over a structure and their terminals are connected to a network of horizontal and vertical conductors.

2. Pulsing Lightning Rods.

- These devices apply a pulsed high voltage to an air terminals tip which gives positive ground currents going upwards meeting descending negative leader strokes.

3. Meshed Cage Lightning Conductors.

- Consist of mesh used across the covering of roofs and side walls of structures.

4. Faraday Cage.

- A container that anchors electromagnetic radiation. It distributes the charge or radiation around the cage's exterior and cancels out electric charges or radiations within the cages interior.

5. Early Streamer Emission (ESE)

- A terminal that is mounted on the top of a structure especially industrial sites, public buildings and sports arenas.

6. Surge Protection devices. (S.P.D)

- S.P.D is used for protection of electrical power supply network and communication systems against over-voltage.

Types of Lightning Arresters.

⇒ A Lightning arrester is a device used to protect a variety of electrical equipment and systems from effects of Lightning.

- Choice of Lightning arrester depends on factors such as;-
  - Voltage.
  - Current.
  - Reliability.
  - Space availability.

1. Rod Gap Arrester
- A gap exist between the end of the rods so that it get filled with air which ionizes and sparks passing the fault current to earth.

2. Sphere Gap Arrester.
- An air gap is provided between two spheres, one of the sphere is grounded, the other one connected to the Line/Supply.

3. Horn Gap Arrester.
- Two horn-shaded pieces of metal separated by a small air gap which ionizes to pass fault current.

4. Multiple Gap Arrester.
- Consist of a series of insulated plates through an air gap. The number of gaps depend on voltage and protects the device through corona discharge.

5. Electrolytic Arrester

- Has high discharge Capacity and operates on the principle of electrolytic cell.

6. Thyrite Lightning Arrester.

- These are used for extremely high Voltage Conditions. Consist of a Ceramic material (Thyrite) which has a Variable resistance.

7. Metal Oxide Lightning Arrester.

- It is also Called Zinc Oxide diverter. They are also Known as gapless Surge diverter. The arrester has a Potential barrier at the edges of Zinc Oxide which Constrains the Passage of current.

END.